

Arman Hassan

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EDUCATION

University of North Carolina at Chapel Hill

Chapel Hill, NC

Bachelor of Science in Computer Science — Minor in Data Science

August 2023 – May 2027

- Coursework: Data Structures, Algorithms, Systems, Computer Organization, Linear Algebra, Models of Computation, Machine Learning, Intro to AI, Mobile Computing, Hardware Security, Data Management

SKILLS

Languages: Python, TypeScript, JavaScript, Go, C++, Java, SQL

Frameworks & Tools: FastAPI, Node.js, REST APIs, React, Next.js, React Native, Tailwind CSS, Playwright

Cloud & DevOps: AWS (S3, IAM), Vercel, Docker, Git, GitHub Actions (CI/CD)

Databases & Storage: PostgreSQL, Redis, pgvector, Amazon S3 Vectors

AI Engineering: OpenAI API, AWS Bedrock, RAG, Embeddings, Tool Calling, LLM Evals, Guardrails

EXPERIENCE

Timing

Remote

Software Engineering Intern

January 2026 – May 2026

- Built a GPT-4o agent at an early-stage AI startup using function calling, structured outputs, and a **RAG pipeline** with OpenAI embeddings and pgvector to rank follow-ups and draft outreach across **1,000+ contacts**
- Created a Manifest V3 Chrome extension using content scripts and browser APIs to integrate Timing's AI networking assistant into LinkedIn and Gmail
- Cut p95 contact API latency by **80%** (450ms to <90ms) using composite PostgreSQL indexes and Redis caching

UNC Chapel Hill

Chapel Hill, NC

Teaching Assistant (COMP 110)

August 2025 – December 2025

- Supported **200+ students** through weekly Python labs and office hours, diagnosing bugs and teaching control flow, functions, OOP, runtime analysis, and memory diagrams
- Reviewed and graded programming assignments, provided code-level feedback, and coordinated with course staff to address recurring student misconceptions

Vogro

Remote

Software Engineering Intern

May 2025 – August 2025

- Partnered with project leads to scope and build REST API integrations (FastAPI, Node.js) connecting a React frontend, backend services, and third-party tools for a platform serving **500+ users**
- Automated data synchronization and cleanup across internal systems with Python scripts, deduplicating and validating records to cut manual processing time by **40%**
- Reduced page load time by **20%** by eliminating redundant API requests and refactoring React components for better performance on lower-end devices

PROJECTS

CourseCompass AI | [Live Demo](#) | [Code](#)

Chapel Hill, NC

Personal Project | Next.js, TypeScript, AWS Bedrock, S3 Vectors, GitHub Actions, Vercel

August 2026

- Shipped a degree-planning assistant covering 200+ UNC programs and tracks with cited answers
- Parsed catalog pages into structured records and indexed Titan embeddings in **S3 Vectors**
- Developed a **RAG pipeline** with **AWS Bedrock**, Claude fallback routing, metadata-filtered retrieval, and citation guardrails to map courses from chat or photo/PDF uploads to remaining requirements
- Set up **CI/CD** with **GitHub Actions** running lint, type checks, and 80+ offline tests, plus Vercel deploys
- Wrote an 18-case live **LLM eval suite**, raising pass rate from an **11%** baseline to **100%**

Jev 2048 | [Live Demo](#) | [Code](#)

Chapel Hill, NC

Personal Project | Next.js, TypeScript, TypeSafe Jev, C++, Python, Oracle Cloud, Vercel

September 2026

- Built a controlled experiment implementing the general-purpose **TypeSafe Jev classifier** as a sequential decision policy to autonomously play 2048
- Benchmarked **five policies** using raw boards, structural features, expectimax, and pretrained n-tuple values
- Deployed a Vercel UI and authenticated Oracle Cloud API serving an MIT-licensed, **67.1M-weight C++ model**
- Five-seed pilot: mean score rose from **796** raw to **10,770** with features and **34,276** with expectimax guidance
- Processed **77,928 API requests** and **133.1M tokens** for **\$5.45**, demonstrating low-cost classification at scale

Arcade-Style Space Battle Game | [Code](#)

Chapel Hill, NC

Hack110 Hackathon — 1st Place | Python, Pygame, OOP, Git

April 2024

- Won **1st place** at Hack110 by shipping a full Pygame arcade shooter end-to-end in under 24 hours
- Engineered core game systems including collision detection, enemy spawning, and progressive difficulty scaling